SPECULATIVE DECODING ACROSS CONSUMER SYSTEMS: GAINS, FAILURES, AND RUNTIME COSTS

Anonymous Authors[1]

ABSTRACT

Fast drafting and cheap target batching need not yield faster speculative decoding once output preservation and state management are included. We examine this gap in four retrospective consumer-system case studies with campaign-specific output comparators. Path controls show that correctness qualification can exclude cheap verification paths, while direct timing shows that an isolated NPU draft advantage does not survive the tested 358H stateful interface. Separate M4 studies retain both profitable and unprofitable outcomes without isolating workload from session effects. We then examine two runtime decisions in separately frozen studies: qualification stops a 358H intervention before performance evaluation, and a one-way M4 guard uses complete realized cost to decide whether to continue speculation. On 60 held-out prompts, Guard achieves 1.3114× post-prefill speedup over target-only (95% prompt-bootstrap CI [1.2487, 1.3787]); Guard and fixed speculation each preserve all 180/180 matched generated-token sequences. Fixed speculation has the higher aggregate speedup estimate, 1.3469×. The contribution is measured evidence of how output qualification and stateful execution constrain usable speculation, with a prospective test of continuation control on one qualified path; no advantage of adding control is established.

1 INTRODUCTION

Speculative decoding promises lower serial decoding latency by using a cheaper draft model to propose tokens that a target model validates together (Leviathan et al., 2023). High acceptance, fast drafting, and cheap batched target execution can each make an implementation look promising. We ask what prevents those local indicators from translating into gains in integrated post-prefill speculative decode on the tested consumer configurations, and whether the measured constraints can guide runtime decisions.

We investigate MacBook Air M4, Core Ultra 9 275HX with RTX 5070 Ti Laptop, Core i5-13400 with Arc B570, and Core Ultra X7 358H with Arc B390 and CPU/NPU drafting. Outcomes differ sharply even within a platform. The representative M4 workload passes its recorded text-path comparator but reaches only 0.842× speedup; a separately preregistered HumanEval workload on the same M4 also passes that comparator and reaches 1.387×. Differing session conditions prevent attributing this contrast to workload alone. Profitability is not a simple hardware capability in these studies.

Two diagnoses explain important failures elsewhere. On RTX 5070 Ti Laptop, correctness qualification removes cheap batching; the tested serialized controls pass the text-path comparator but cost more. B570 serialization improves agreement without qualifying the full workload. On 358H, the isolated NPU draft advantage does not survive the tested stateful integration, while complete target-side work dominates CPU drafting. These diagnoses neither exhaust possible causes nor explain the M4 contrast. Hardware, workload, and runtime co-vary; all gains concern post-prefill decode, excluding loading and initial prompt prefill.

Our central contribution is a measured account of how the output contract and the actual stateful interface change which speculative execution opportunities remain usable. Path controls expose the cost of losing cheap verification; direct timing exposes work absent from isolated draft measurements. Separately frozen runtime studies test the resulting decisions: 358H has no qualified intervention to time, while an M4 guard tests whether a simple complete-cost continuation rule can execute exactly and profitably on held-out requests. It does, but fixed speculation has the higher aggregate speedup point estimate. The contribution is empirical and configuration-scoped. Heterogeneous placement, cost/commit accounting, deterministic inference, and adaptive speculation are established (Zhang et al., 2025; Lv et al., 2025; Liu et al., 2026; Kim et al., 2026; Gond et al., 2026; Liu et al., 2025); the threshold policy is not a new speculative algorithm.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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2 Experimental Question and Measurement Discipline

We investigate the gap between favorable component measurements and integrated post-prefill decode outcomes. Acceptance describes candidate agreement, isolated draft throughput describes one execution boundary, and target-forward batching describes another. None alone measures the complete speculative loop. We use an output contract and an accounting lens to make those boundaries explicit before comparing the frozen campaigns.

2.1 A path-qualified output contract

A greedy speculative round proposes a draft sequence, verifies it with the target, and commits the matching prefix followed by a correction or bonus token when the remaining output budget permits. Rejected candidate state must be removed or reconstructed consistently. Our intended output contract requires generated token IDs to equal the matched target-only greedy output under the frozen configuration. This makes output preservation an explicit requirement and compares latency for the same generated sequence. It is an empirical property of a model, runtime, execution path, and workload, not a property inferred from the algorithm name.

The recorded comparators are not uniform. Historical M4, RTX, and B570 campaigns compare token arrays reconstructed from parsed CLI text; their reported equality counts establish agreement through that text-processing path, not direct generated-ID equality. The 358H runner and prospective M4 runtime study compare generated IDs directly. We retain the historical counts and evidence classes, but do not promote text-path agreement to proof of the stricter ID contract.

We call a path *admissible* only within its tested workload and recorded comparator. For historical text-path comparisons this denotes campaign qualification, not certification under the direct-ID contract. A failed pair blocks a workload-global optimization claim even if the timing is favorable. A small qualification probe establishes eligibility for broader testing, not universal equivalence. We retain blocked-path timings as diagnostics with their correctness status visible. This deliberate contract is stricter than an approximate-quality deployment objective; HumanEval token equality does not establish functional correctness or pass@k.

2.2 Evidence classes

Primary results answer a frozen workload question after its correctness gate. **Secondary** results are registered variants or explicitly bounded follow-up controls. **Mechanism-supported** claims use path controls or timed regions. **Diagnostic** values describe correctness-blocked or otherwise restricted paths. These labels describe permitted inference, not just measurement precision. In particular, the text-path-qualified RTX serialized controls are secondary evidence; they do not replace its blocked primary fast path. B570 follow-ups remain diagnostic after the formal stop.

2.3 Complete round costs

Let T_b be the matched target-only cost per output token. The realized draft time T_d^{real} denotes the realized draft-subsystem time per round, including any prefix replay performed by its interface. The complete target time $T_{\text{target,round}}^{\text{real}}$ includes all measured target-side round work: candidate evaluation and any correction, bonus, or rollback work inside that region. Residual synchronization, transfer, orchestration, and bookkeeping time not included in either region is T_x. Let C count *all* committed output tokens. With consistent timing and aggregation boundaries,

$$S = \frac{T_b C}{T_d^{\text{real}} + T_{\text{target,round}}^{\text{real}} + T_x} \tag{1}$$

$$= \frac{C}{D_{\text{real}} + R_{\text{target}} + X}, \tag{2}$$

where each denominator term is normalized by T_b. Break-even requires $C > D_{\text{real}} + R_{\text{target}} + X$. These identities organize measurements; they are not a new profitability theory. Cost/commit decompositions already appear in runtime-controlled studies (Kim et al., 2026).

At an aggregate level, round means and matched total-time/token accounting must use the same weights. A ratio of total times need not equal a geometric mean of prompt speedups. Likewise, accepted draft tokens per round omit correction or bonus output and cannot automatically be used as C. We do not infer unmeasured replay-only, transfer-only, or correction-only components by subtraction from incompatible experiments.

2.4 A separate target-forward proxy

For context n and path e, define

$$V_k^{\text{micro}}(n, e) = \frac{T_{\text{target forward}}(k, n, e)}{T_{\text{target decode}}(1, n)}. \tag{3}$$

This microbenchmark asks whether a target can process a causal block near the cost of a single decode. It omits draft generation and can omit target correction/bonus and state-management work. Consequently V_k^{micro} is not automatically R_{target}. Likewise, isolated draft throughput cannot substitute for realized stateful draft cost. These distinctions guide diagnosis after the observed outcomes; they do not supply a universal explanation for those outcomes.

Table 1. Frozen platform and runtime scope. All final target models are Qwen3-8B Q8_0. Greedy execution is compared with matched target-only references; historical M4/RTX/B570 use re-tokenized text, while 358H uses direct IDs. Hardware and software co-vary.

System	Runtime / target backend	Draft model / placement	Evidence scope
MacBook Air M4	llama.cpp b10809 / Metal	Qwen3-0.6B Q8_0 / Metal	Representative 24Q and separate HumanEval30
Core Ultra 9 275HX + RTX 5070 Ti Laptop	llama.cpp b10839 / CUDA	Qwen3-0.6B Q8_0 / CPU or same GPU	Blocked fast path; text-path-qualified controls
Core i5-13400 + Arc B570	llama.cpp b10839 / Vulkan0	Qwen3-0.6B Q8_0 / B570	Formal gate failure; later diagnostics
Core Ultra X7 358H + Arc B390	llama.cpp b10839 / Vulkan0; OpenVINO draft	Qwen3-0.6B symmetric INT4 / CPU or NPU	Closed Phase 3; direct stateful decomposition

3 Experimental Methodology and Platforms

3.1 Frozen systems and workloads

The cross-platform hardware synthesis is retrospective: campaign-level primary outcomes were frozen independently. Later path controls and mechanism analyses retain their stated secondary, mechanism-supported, or diagnostic scope; they do not form a jointly preregistered cross-platform experiment.

Table 1 summarizes the completed hardware campaigns. The separately frozen runtime protocols and their distinct M4 estimator appear in Section 7. Final comparisons process one request at a time (sequence batch one). The final target is Qwen3-8B Q8_0; the native draft is Qwen3-0.6B Q8_0. The 358H heterogeneous adapter instead uses the qualified symmetric INT4 OpenVINO draft. We retain this quantization difference rather than describing the stacks as identical. Earlier M4 MLX-LM precision/correctness failures belong to qualification history and are distinct from the final llama.cpp Metal path.

The M4 representative sample freezes 24 previously unseen Spec-Bench prompts (Xia et al., 2024), with four from each of six workload groups and prior diagnostic IDs excluded. It uses three randomized paired repetitions. A separate preregistration freezes 30 HumanEval IDs (Chen et al., 2021) before prompt download or inspection; fixed $k = 2$ is the primary condition. The registered confidence-threshold conditions remain secondary. The subsequent HumanEval comparisons retain the frozen code-workload scope, three paired repetitions, greedy decoding, and a 128-token cap. They do not execute generated code.

Primary timing compares matched decode regions, excluding model loading and initial prompt prefill. In the native campaigns we use the frozen runtime decode counters; 358H explicitly times the complete post-prefill speculative loop. Replaying the committed prefix *inside* that loop remains charged to drafting. Thus the measurements are conditional decode costs, not whole-request latency including startup. Target-forward controls have their own narrower timing boundary. Different runtime versions, quantization, and device paths prevent a hardware-only causal comparison.

3.2 Statistical estimators and uncertainty

For each hardware-study prompt we reduce its three paired baseline/speculative ratios, then report the geometric mean across prompts. M4 and RTX use a within-prompt geometric mean, as does the B570 diagnostic analysis; 358H uses a within-prompt median. Intervals are 95% prompt bootstraps. M4 HumanEval uses 100,000 resamples; the other reported campaigns use 10,000. Frozen seeds and endpoint conventions are given in the separate appendix. We preserve campaign-specific estimators and finite-bootstrap input order rather than harmonizing them after observing outcomes.

These intervals quantify prompt sampling conditional on the campaign. They do not encompass all session, temperature, or runtime-state uncertainty. Randomized adjacent pairing reduces drift but does not remove it. We therefore keep the representative and code-conditioned M4 studies separate, including their differing session conditions.

3.3 Provenance and energy boundaries

Frozen definitions, processed summaries, scripts, and archived evidence anchor every quantitative claim in an internal ledger. Equality counts are interpreted using the comparator distinction in Section 2, not as uniform direct-ID evidence. Windows raw per-run provenance is weaker than the other completed platforms, although committed pair tables and mechanism captures support recomputation. The 358H reconstructed draft source is `STRONGLY_SUPPORTED`; exact historical upstream identity is not proven. The header-only blocked-preflight derivative is excluded from final analysis.

Energy is interpreted only within its qualified sensor bound-

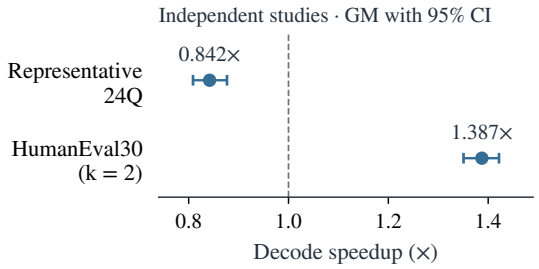


Figure 1. **Opposite profitability regimes on the same tested M4 stack**. Representative 24Q yields 0.842× [0.809, 0.877], while the separately preregistered HumanEval30 fixed-$k = 2$ study yields 1.387× [1.351, 1.422]. Points are campaign prompt geometric means of paired decode speedups; intervals are campaign-specific 95% prompt-bootstrap intervals. The dashed line marks parity. Both pass the recorded re-tokenized-text comparator (72/72 and 90/90 pairs); these are not direct generated-ID comparisons or functional HumanEval pass@k. These are separate studies, not a paired between-workload comparison; differing workloads and session conditions do not isolate workload-only causality.

ary. M4 measures package aggregate excluding separate DRAM; B570 measures adapter board energy. RTX board and CPU-package domains are not a common whole-system boundary, and its serialized HumanEval follow-up lacks an admissible formal energy comparison. No end-to-end Phase-3 energy claim is made for 358H. We do not rank platform energy efficiency.

4 Gains Do Not Translate Uniformly

The frozen outcomes establish the cases from which we examine two measured failure boundaries. Table 2 keeps correctness and evidence class beside speed: a blocked path is a diagnostic observation, not a slower admissible optimization.

4.1 M4: outcomes in two separate studies

The representative frozen M4 sample produces 0.842× speedup (95% CI [0.809, 0.877]), with 72/72 text-path-equal pairs and only 1/24 prompts faster. The separately preregistered HumanEval fixed-$k = 2$ study produces 1.387× [1.351, 1.422], with 90/90 text-path-equal pairs and 30/30 prompts faster. Figure 1 retains both; the positive code-conditioned result does not replace or pool with the negative representative result.

The same M4 supports both profitable and unprofitable observed regimes that pass the historical text-path comparator. Profitability is therefore not a simple hardware capability property in these studies. These outcomes do not establish general M4 profitability or isolate unified memory as a cause of the profitable result. These are workload-conditioned observed regimes, not a controlled estimate of workload causality: workloads and session conditions differ. The earlier discovery effect size is not promoted as a confirmatory estimate. Historical MLX failures also do not negate the text-path-qualified historical Q8_0 Metal result.

For primary fixed-$k = 2$ HumanEval, the package-efficiency ratio (baseline/speculative J per output token) is 1.013× [0.990, 1.034]. The interval includes parity; a package-energy improvement is not established. The registered $p_{\min} = 0.9$ condition has secondary energy evidence, discussed separately in the appendix. The preregistered corrected synthetic ceiling was not executed. We retain that deviation rather than treating the earlier synthetic diagnostic as its replacement.

4.2 Across stacks: negative and blocked outcomes

On RTX 5070 Ti Laptop, the serialized CPU-to-RTX control reaches 0.404× [0.390, 0.418], while the same-GPU draft control reaches 0.650× [0.629, 0.673]. Both pass 90/90 text-path comparisons and accelerate zero of 30 prompts. The higher speed of same-GPU placement shows that placement and associated integration costs matter, but its continued loss rules out transfer as the sole explanation.

On Arc B570, neither the fast path nor the serialized fallback supports a workload-global admissible optimization result. The latter's diagnostic speedup is 0.3087× [0.2927, 0.3246], with zero of 30 prompts faster. That interval describes timing on a partially mismatching path; its precision does not turn it into an admissible speed claim. The formal failure and stop rule remain the platform result. Board-energy observations likewise remain diagnostic and boundary-limited.

On Core Ultra X7 358H, the closed Phase-3 heterogeneous runner is exact on both CPU and NPU drafting: 90/90 pairs per path, with zero of 30 prompts faster. CPU achieves 0.734× [0.693, 0.771]; NPU achieves 0.352× [0.333, 0.369]. These exact negative outcomes coexist with favorable isolated draft measurements. Section 5 investigates which measured boundaries account for the failed local promise.

5 From Local Opportunity to Realized Gain

The observed outcomes motivate two diagnoses supported by path controls and direct timing. Figure 2 locates the execution-path qualification and stateful-realization boundaries. These diagnoses apply to identified configurations and do not constitute an exhaustive theory. The M4 contrast remains an independent empirical observation; neither diagnosis is assigned to it.

Table 2. Headline decode outcomes. Speedup is baseline/speculative time, aggregated by the campaign's frozen prompt estimator. Brackets are 95% prompt-bootstrap CIs. Matching pairs and faster prompts are separate metrics. M4/RTX/B570 count re-tokenized-text agreement; 358H counts direct-ID agreement. B570 timing is diagnostic, despite a reported CI.

Path / workload	Speedup [95% CI]	Matching pairs	Faster prompts	Class
M4 representative 24Q	0.842 [0.809, 0.877]	72/72	1/24	Primary
M4 HumanEval fixed-$k = 2$	1.387 [1.351, 1.422]	90/90	30/30	Primary
CPU to RTX 5070 Ti Laptop	0.404 [0.390, 0.418]	90/90	0/30	Second.
RTX 5070 Ti Laptop to itself	0.650 [0.629, 0.673]	90/90	0/30	Second.
B570 serialized fallback	0.3087 [0.2927, 0.3246]	60/90	0/30	Diag.
358H CPU draft	0.734 [0.693, 0.771]	90/90	0/30	Primary
358H NPU draft	0.352 [0.333, 0.369]	90/90	0/30	Primary

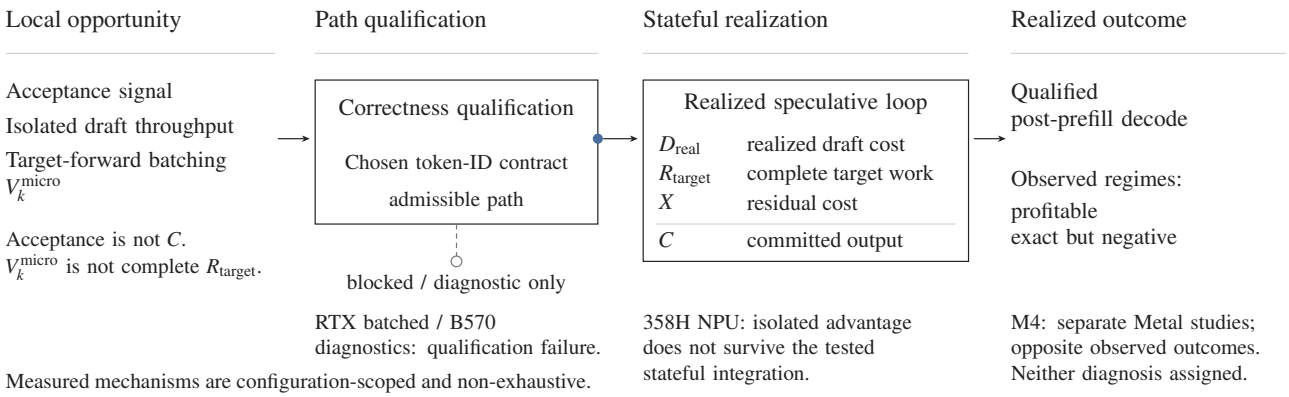


Figure 2. **From local opportunity to integrated post-prefill decode outcomes.** Local acceptance, isolated drafting, and target-forward batching do not by themselves establish integrated gains. Arrows mark evaluation boundaries, not a guarantee of gain. The output requirement excludes the tested RTX 5070 Ti Laptop batched path; B570 serialization improves agreement without qualifying the full workload. Blocked timings remain diagnostic. For the qualified 358H NPU path, full-prefix replay and proposal share one synchronous OpenVINO call, and the isolated draft advantage does not survive the tested stateful integration. D_{real}, R_{target}, and X normalize measured draft, complete target, and residual costs by matched target-only token cost; C counts all committed output, including correction/bonus tokens. M4's separate frozen studies yield opposite profitability regimes with text-path agreement. Historical M4/RTX/B570 equality labels refer to that comparator, not direct-ID certification. Workload and session conditions differ, and neither diagnosis is assigned to the contrast. The measured mechanisms are configuration-scoped and non-exhaustive; timing excludes loading and initial prompt prefill.

5.1 Execution-path qualification

Cheap physical batching is not sufficient. The historical M4 Metal path combines frozen-workload text-path agreement with a target-forward proxy near one baseline step. Across its tested contexts, V_2^{micro} is approximately 1.02–1.08. This is a shared-path reference from a separate micro study, not evidence for either diagnosed mechanism or a complete round decomposition. It does not explain the M4 outcome contrast.

On RTX 5070 Ti Laptop, the batched CUDA target also has near-one-step two-token forward cost: $V_2^{\text{micro}} = 1.009$ and 0.983 at the two measured depths. The intended fast HumanEval execution nevertheless fails the frozen text-path comparison gate. Its timings cannot support an admissible optimization claim. The serialized control changes the micro-batch to one and raises the measured proxy to 2.431 and 2.137; the full frozen control workload passes its text-path comparator. Figure 3 therefore relates a physical opportunity to its empirically qualified status, rather than ranking GPUs.

These controls are deliberately secondary. They qualify a slower path after the native fast attempt failed; they are not a rescued primary success. The verification microbenchmark explains lost batching opportunity, while the post-prefill decode controls include drafting and integration. We cannot recover a full R_{target} decomposition by substituting the forward-only ratio into Equation 2.

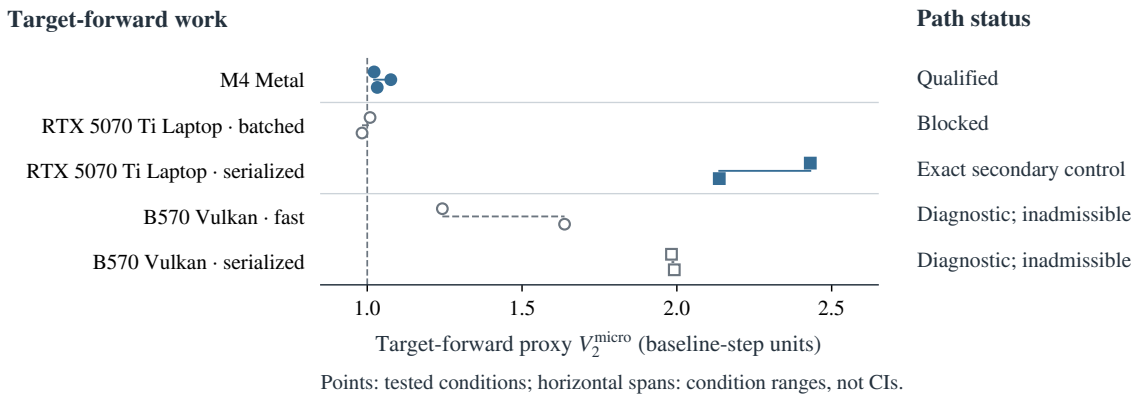


Figure 3. **Correctness qualification changes which verification path is usable**. Each mark is one measured context cell; range segments span those cells, not confidence intervals. Filled markers indicate paths qualified by the historical text-path comparator; open markers indicate blocked or diagnostic paths. Serialization yields a text-path-qualified secondary RTX control, but the B570 serialized diagnostic remains globally inadmissible despite improving text-path agreement from 45/90 to 60/90 pairs. Contexts and runtime stacks are not fully matched. The target-forward proxy V_2^{micro} is not complete realized target-side cost R_{target} or an end-to-end speedup. Qualification is scoped to the frozen workload and execution path, not a device family. M4 contexts are 512/2,048/8,192 tokens, RTX depths 0/256, and B570 contexts 512/2,048; vertical offsets separate nearby cells without changing horizontal values. Circles denote batched/fast paths and squares serialized paths. M4 cells provide a shared-path reference; neither diagnosed mechanism is assigned to the M4 outcome contrast. The separate appendix preserves the context-cell lookup.

A bounded numerical divergence. The frozen CUDA reproducer repeats the HumanEval/106 divergence in three of three runs at the same captured location. Sequential and serialized execution select token 2790, whereas batched execution selects 2629. A separate same-commit diagnostic build captures the corresponding score-margin change from $+0.080948$ to -0.126457. That numerical instrumentation does not replace the original formal binary or its measurements.

The evidence supports stable batch-shape-dependent greedy argmax divergence in this llama.cpp/CUDA/Q8_0 configuration. It does not identify a universal CUDA failure or a specific responsible kernel. Floating-point and batch-dependent output differences are established research topics (Yuan et al., 2025; Gond et al., 2026); the systems consequence here is that the cheap path is removed from the set admitted by the study's contract.

Serialization does not establish exactness. B570 provides a complementary boundary. Its formal v2 correctness gate fails, so the preregistered stop rule prevents a formal verification/performance claim from that execution. Later fast-path diagnostics yield 45/90 text-path-equal pairs. The unique serialized fallback improves this to 60/90, with 20/30 wholly matching prompts under that comparator, but still fails workload-global equality. Its V_2^{micro} is approximately 1.983–1.992 and remains diagnostic.

A smaller serialized probe had passed before the full workload exposed mismatches. Passing such a gate is evidence for its tested scope, not proof of broader correctness. We therefore use "serialized V_2^{micro}" or "V2_serialized" for this B570 path. An execution property must not be renamed as an exactness guarantee.

This negative control matters because it prevents an attractive simplification: selecting serialized verification is not by itself a sufficient correctness policy. Qualification must cover the actual path and intended workload. Text-path-matching subsets may be descriptively useful, but conditioning on those subsets after the run cannot establish global performance.

5.2 Stateful realization

On Core Ultra X7 358H, the isolated NPU draft has higher median steady-state throughput than CPU at the tested contexts, with a larger advantage at longer contexts. This local indicator does not survive integration: both qualified paths are exact but unprofitable, and the NPU path is slower. The measured regions reveal different costs on the two paths. The direct round records show mean committed progress $C = 2.4631$ and 2.4584, respectively, approximately 2.46. This is complete committed output, not the accepted-draft counter used in older native records.

The target maintains persistent state, evaluates proposals, removes invalid candidate state when needed, and decodes correction or bonus output. Its complete measured region accounts for 87.34% of CPU-path speculative time. The NPU

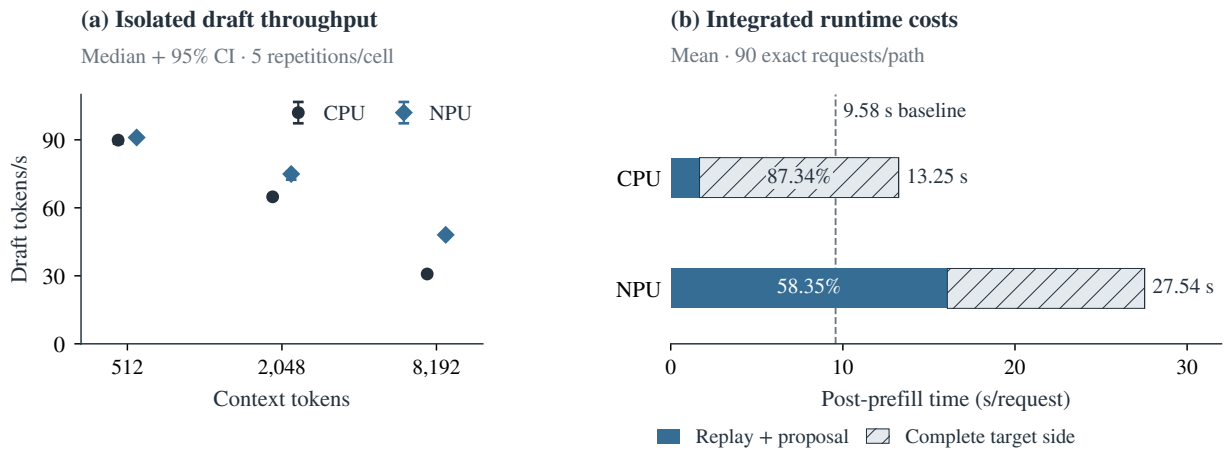


Figure 4. **Isolated draft advantage does not survive the tested stateful integration**. (A) Phase-1 median steady-state draft throughput and 95% repetition-bootstrap intervals, five repetitions per cell; the NPU median advantage grows across the tested contexts. (B) Phase-3 arithmetic mean post-prefill seconds per request over 90 observations per path, all exact. CPU and NPU totals are 13.25 s and 27.54 s against a 9.58 s mean baseline. The CPU path is target-side dominated (87.34%); the NPU path spends 58.35% in the measured replay-plus-proposal call. The 0.734× and 0.352× headlines use the separate prompt-GM estimator and are not ratios of the arithmetic mean bars. Replay and proposal share one timed OpenVINO call; replay-only time is not measured. Initial prompt prefill is excluded. This diagnoses the tested runtime interface, not intrinsic NPU capability or a measured benefit from a state-persistent intervention. CPU replay plus proposal is 12.63%; NPU complete target work is 41.63%. Residuals are 4.81 ms (0.04%) and 5.06 ms (0.02%). Prompt-GM speedup 95% CIs are [0.693, 0.771] and [0.333, 0.369], respectively. The vertical dashed line is the shared arithmetic-mean baseline. Isolated throughput excludes the full-prefix state cost; device marks are horizontally offset for readability.

path instead spends 58.35% in the synchronous OpenVINO call that replays the full committed prefix and generates a proposal. Replay and proposal are inseparable within this measurement; we do not invent a replay-only value. Residual orchestration occupies only 0.04% and 0.02%, so it cannot explain the main loss.

Figure 4 separates isolated steady-state throughput from this realized request decomposition. The isolated NPU draft can outperform the CPU at the tested contexts, but this advantage does not survive the qualified state interface. The conclusion concerns that implementation, not insufficient NPU compute capability. The subsequent runtime study tests intervention eligibility separately; these Phase-3 timings do not estimate a persistence benefit.

Using arithmetic totals at a common boundary reconstructs Equation 2. CPU has $(D_{\text{real}}, R_{\text{target}}, X) = (0.4304, 2.9767, 0.00124)$ and aggregate speedup $0.7227\times$; NPU has $(4.1251, 2.9435, 0.00130)$ and $0.3477\times$. These agree with the ratios of total baseline/speculative times. They intentionally differ from the primary prompt-level geometric means in Table 2.

The earlier Phase-2 screen combined isolated drafting and a target-forward proxy under favorable residual-cost assumptions. It was an eligibility screen, not a validated post-prefill decode prediction. Phase 3 shows why those proxies must be replaced with complete target-side work and realized stateful draft cost. No end-to-end Phase-3 energy result is claimed, and source reconstruction retains its `STRONGLY_SUPPORTED` qualification.

6 From Measured Constraints to Runtime Decisions

The path controls and stateful measurements identify two ways component promise can fail to describe a usable optimization. The cheapest measured verification path may not satisfy the output contract. The fastest isolated draft device may execute additional work through its state interface. Lowering draft cost can also leave complete target-side work dominant, as in the CPU-draft 358H path. These are consequences of the tested executions, not hardware ranks or an exhaustive performance theory.

The resulting runtime questions are whether a candidate path qualifies, and whether its complete realized cost warrants continued speculation. Qualification alone does not establish profitability; the separate M4 campaigns motivate measuring the latter without identifying a workload-only cause. The following studies examine these two decisions on

different stacks. They do not form an intervention/ablation chain or estimate a benefit from persistent draft state.

7 Qualification and Realized-Cost Runtime Control

We examine qualification and continuation in separately frozen runtime studies. The 358H candidate stops at qualification. On M4, an available exact path permits a prospective test of a simple cost rule: Guard remains faster than target-only, but fixed speculation has the higher aggregate speedup point estimate. These are complementary tests of the decisions in Section 6, not two successful optimization interventions.

7.1 An intervention stopped by qualification

The 358H component measurements motivated persistent draft state and a wider target execution setting. The implementation ran the target and draft backends in separate processes. Synchronous draft calls, including communication and state work, remained inside the target worker's post-prefill timer. The executable implementation still had to pass correctness qualification. CPU persistent-state restore/reconcile correctness could not be established under the tested representation; the Optimum NPU state-capable path was unavailable with the frozen dynamic-shape IR. These are implementation boundaries, not evidence that the devices cannot retain state.

The target setting `n_ubatch=2` (UB2) passed the small gate but failed full qualification. Correctness-only follow-up controls on Qwen3-8B Q8_0, llama.cpp b10839, and Vulkan0 reproduced different greedy tokens under UB1 and UB2 even with identical exact prefix tokens. Request-order and separate-process controls reproduced the target-only divergence. Replaying frozen proposals also reproduced the speculative divergence without live NPU draft inference. The latter depended on earlier target execution history; the controls found no position drift or specific rollback-boundary bug. The responsible floating-point or backend operation remains unidentified.

Under this frozen tested target stack, UB2 did not preserve the strict exact-output contract. The formal registry retained only the target reference, so the runtime study had no qualified intervention pair to evaluate. No formal R2 performance run occurred, and the follow-up diagnostics are invalid for performance claims. Qualification excluded an intervention motivated by the measured bottlenecks before its speed was evaluated.

7.2 A prospective cost guard on M4

The M4 experiment asks whether a frozen rule based on already-realized complete cost can control continuation while preserving exact output and profitable execution against target-only. It does not assume that adding control improves on fixed speculation. We retain the Qwen3-8B/0.6B Q8_0 pair on llama.cpp Metal, fixed $k = 2$, greedy generation, and a 128-token cap. Guard begins on the same speculative path as Always-Spec and can fall back permanently to target-only within a request. It changes neither proposal width nor device placement.

At each eligible completed round, Guard divides all elapsed charged generation cost by committed output progress C. This count includes the initial target seed and committed correction or bonus tokens, but excludes rejected proposals. Guard continues if this cumulative cost per token is at most $(1 - m)R_{\text{ref}}$, a target-only reference frozen from calibration; otherwise it falls back. If a decision is reached with zero committed progress, Guard also falls back. Before each control decision, invalid candidate suffix state is removed from the target and draft contexts, preserving committed history for target-only continuation. The enclosing post-prefill clock includes generation-loop initialization, seed-token sampling, drafting, target work, synchronization, state reconciliation, control, and transition costs. Model loading, sampler/speculator setup, initial prompt prefill, and the speculative begin hook precede the timer; the result is not whole-request latency.

The reference is the median across calibration prompts of each prompt's median target latency divided by its median output-token count. The preregistered candidate grid varies the first decision round and margin. Selection minimizes calibration geometric-mean latency relative to the offline minimum of target-only and Always-Spec among exact candidates, with frozen tie-breaks. The selected rule uses margin 0 and first decides after 8 completed speculative rounds; it then decides after each later round while speculation continues. A request ending earlier makes no decision. The online policy receives neither task/family labels nor future output length, current-request target-only latency, or oracle outcomes.

7.3 Held-out evaluation

Historical M4 outcomes informed the study design. Prospective here means that the policy was frozen before the reserved prompt outcomes were measured, not that the design was uninformed by earlier campaigns. IDs marked as previously executed or outcome-exposed in the frozen inventory were excluded before deterministic hash-ranked selection. Disjoint splits contain 40 calibration prompts, 16 qualification prompts, and 60 held-out prompts, with equal HumanEval and Spec-Bench counts within each split. The

Table 3. M4 per-prompt median post-prefill latencies. Each speculative condition is 180/180 direct-token exact. The first two rows report speedups. The offline oracle uses the lower target-only or Always-Spec median and is unavailable online; its row reports latency regret. The separated Guard/Always-Spec contrast is reviewer-requested post-study paired secondary analysis; values above one mean Guard is slower.

Latency ratio	GM	95% prompt CI
Target / Guard	1.3114	[1.2487, 1.3787]
Target / Always-Spec	1.3469	[1.2867, 1.4088]
Guard / offline oracle	1.0278	[1.0137, 1.0446]
Post-study paired secondary latency ratio		
Guard / Always-Spec	1.0271	[1.0130, 1.0436]

policy and reference were frozen before qualification and formal measurement; the appendix documents the audited recovery of a freeze-hash metadata defect. Both Always-Spec and Guard passed 48/48 direct-token comparisons in qualification.

The formal study runs 3 repetitions of target-only, Always-Spec, and Guard on each held-out prompt, randomizing condition order within prompt and repetition: 540 observations. Both speculative conditions are 180/180 direct-token exact, with zero mismatches. For each prompt and condition, analysis takes the median latency across repetitions, then the geometric mean of prompt latency ratios. This ratio-of-medians estimator differs from the historical M4 hardware estimator. Confidence intervals use 10,000 prompt bootstrap resamples with seed 20260924; repetitions are not independent sampling units.

Guard is profitable against target-only on this held-out workload (Table 3). Fixed speculation has the higher aggregate speedup point estimate. A reviewer-requested post-study paired secondary analysis of the frozen prompt medians gives Guard/Always-Spec latency 1.0271 [1.0130, 1.0436]. The interval lies above parity: Guard had higher latency on this comparison. By prompt median, Guard is faster on 24 prompts, Always-Spec on 36, with 0 ties. Guard falls back in 30.6% of its 180 request executions; this measures control activity, not the fraction of beneficial decisions. Guard's geometric-mean latency relative to the offline oracle is 1.0278×. That oracle chooses the lower target-only or Always-Spec median after measurement and was unavailable to the runtime. It is a descriptive hindsight comparator, not an achievable-policy guarantee.

The result demonstrates exact, profitable prospective execution of a complete-cost continuation rule. Neither fallback activity nor the oracle ratio establishes an incremental benefit of control, risk reduction, or threshold optimality. The experiment supplies no evidence of policy transfer to another stack. Small individual Spec-Bench strata do not support stable subgroup estimates. No runtime energy claim is made. The separate appendix documents the archived recomputation and audited freeze-hash metadata exception.

8 Related Work

Performance and cost accounting. Speculative decoding amortizes target computation while preserving a target sampling distribution under its algorithmic assumptions (Leviathan et al., 2023). Liu et al. (2026) systematically study speculative methods in vLLM, including verification dominance and workload/batch effects. Kim et al. (2026) control shared runtime policies and express speedup through committed work and conditional execution cost. Lyu et al. (2026) derive profitability conditions for edge–cloud execution. TurboSpec profiles execution and uses feedback to adjust or disable speculation for serving goodput (Liu et al., 2025). BanditSpec selects speculative configurations online and analyzes stopping-time regret (Hou et al., 2025); our oracle ratio is a descriptive latency comparison, not a regret guarantee. Our retrospective case studies examine two measured boundaries of integrated post-prefill decode: which verification paths remain usable after correctness qualification, and whether isolated draft throughput carries through the actual state interface. Our prospective M4 study adds a frozen one-way control experiment on a correctness-qualified consumer path. It is not a comparison against serving controllers or a claim that adaptive speculation or cost/commit accounting is new.

Heterogeneous execution. Dovetail places drafting on a GPU and target verification on a CPU (Zhang et al., 2025); DuoDecoding studies CPU/GPU placement with hardware-aware draft budgets (Lv et al., 2025). Mirror-SD explicitly considers GPU/NPU mapping and complementary speculative pipelines (Bhendawade et al., 2025). Heterogeneous speculative execution is therefore established prior art. Our negative heterogeneous outcome characterizes the qualified interface and does not contradict acceleration under other placements or runtimes.

Correctness and numerical reproducibility. Zhang et al. (2026) analyze batch speculative-decoding synchronization errors and distinguish them from residual numerical mismatches. Their paper revises the earlier *Batch Speculative Decoding Done Right* preprint. Yuan et al. (2025) study numerical sources of output variation across system configurations; LLM-42 uses verified speculation to enforce deterministic inference (Gond et al., 2026). Our CUDA observation is not a new numerical phenomenon, and we do not equate it with ragged-batch synchronization bugs. We measure its consequence for an admissible target path, with B570 showing why serialization alone cannot certify the

intended workload.

State and memory reuse. Draft-side state management also has prior art. Yuan et al. (2026) maintain incrementally updated compressed draft KV memory for long contexts. Such designs differ from the full-prefix replay interface measured here, but preclude treating state reuse itself as a new idea. Our evidence identifies a realized replay-plus-proposal cost in a consumer OpenVINO NPU path. The subsequent 358H runtime study stops at qualification, while M4 evaluates cost control on an available exact path. Neither result establishes a measured benefit from persistent draft state.

9 LIMITATIONS

The four consumer systems use a narrow Qwen3 model family, with hardware, operating systems, runtime versions, backends, and some quantization choices co-varying. The cases establish specific outcomes and within-platform mechanisms, not an architectural ranking, a population failure rate, or controlled portability. The two diagnoses do not causally explain the separate-session M4 contrast. Serving concurrency, production load, and broader model families are outside the measured scope.

The strict output contract is conditional on the frozen reference; another quality objective could admit other paths. Historical M4, RTX, and B570 equality counts use re-tokenized CLI text, with campaign-specific parsing. They cannot establish direct generated-ID equality; the historical labels and counts remain evidence of their recorded comparator only. HumanEval functional execution and pass@k were not evaluated. CUDA/B570 and the 358H runtime qualification findings concern their tested paths, not general backend unsuitability. Replay and proposal share one timed region, so their individual causal costs and a persistence benefit are not established. The 358H intervention did not reach formal performance evaluation.

Campaign estimators and bootstrap conventions differ. Prompt intervals do not cover all session/thermal uncertainty. Timing excludes initial prefill and loading and is not whole-request user latency. Energy domains differ, the corrected M4 synthetic-ceiling preregistration remains unexecuted, and no common energy ranking or end-to-end 358H Phase-3 energy claim is supported.

The runtime result covers one M4 model/runtime pair, 60 held-out prompts, fixed $k = 2$, and one-way fallback. The paired Guard/Always-Spec contrast is post-study secondary analysis, not the registered primary endpoint; its prompt-bootstrap uncertainty excludes session and recalibration variation. Guard had higher latency on this frozen comparison. Policy transfer, stable small-stratum conclusions, and runtime energy benefit remain unestablished. Historical outcomes informed the design; policy freezing protects the reserved prompt evaluation, not ignorance of earlier platform results.

Windows raw per-run provenance is weaker than in the other campaigns. The 358H draft-source reconstruction is `STRONGLY_SUPPORTED`, not proven historical identity, and its runtime diagnostic is available as a compact record without local raw-trace replay. The M4 freeze-hash metadata defect was independently reconciled against the actual freeze, formal-start commit, and archived observations. The erroneous original markers and pipeline stop remain intact; the appendix explains this recovery rather than treating it as uninterrupted pipeline validation.

10 CONCLUSION

The useful speculative path is not necessarily the one suggested by favorable component measurements. In the tested stacks, output qualification can remove cheap verification, and the state interface can consume an isolated draft-device advantage. The runtime studies expose the resulting decisions: the 358H intervention stops at qualification, while M4 permits a prospective test of continued speculation using complete realized cost. That guard preserves exact outputs and remains profitable against target-only, without demonstrating an improvement over fixed speculation. The systems lesson is to distinguish physical opportunity, satisfaction of the intended output contract, and complete execution cost when evaluating a speculative runtime choice.

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